

1. Создайте на сервере для 1 практики ключ ssh при помощи программы ssh-keygen

```
eltex-pg1-v16@eltex-2025-autumn:~$ ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/home/eltex-pg1-v16/.ssh/id_rsa):
Created directory '/home/eltex-pg1-v16/.ssh'.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/eltex-pg1-v16/.ssh/id_rsa
Your public key has been saved in /home/eltex-pg1-v16/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:U60WCteClyATWg92oD59poa6XyBVNRA6qzpjkmNIhfQ eltex-pg1-v16@eltex-2025-autumn
The key's randomart image is:
+---[RSA 3072]-----+
|.B.=+o|
|*.O. |
|o BEo o|
|.O.= o o + .|
|+=+o.OO= S|
|+=+o.++ + .|
|oo o. .|
|=....|
|=....|
|=....|
+-----[SHA256]-----+
```

2. Скопируйте созданный ключ на сервер для 2 практики для пользователя root при помощи программы ssh-copy-id

```
eltex-pg1-v16@eltex-2025-autumn:~$ ssh-copy-id root@172.16.9.190
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/eltex-pg1-v16/.ssh/id_rsa.pub"
The authenticity of host '172.16.9.190 (172.16.9.190)' can't be established.
ED25519 key fingerprint is SHA256:YNVzj+RtetZfXv5XUn/CUjeEcaMD4eg3jyBEuGGIS+o.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter
out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed if you are prompt
ed now it is to install the new keys
root@172.16.9.190's password:
Number of key(s) added: 1
Now try logging into the machine, with: "ssh 'root@172.16.9.190'"
and check to make sure that only the key(s) you wanted were added.
```

3. Подключитесь к серверу для 2 практики под пользователем root и сравните содержимое файла открытого ключа на сервере 1 ~/.ssh/*.pub и файла ~/.ssh/authorized_keys на сервере для 2 практики, а так же права доступа для каждого из файлов

```

eltex-pg1-v16@eltex-2025-autumn:~$ ssh root@172.16.9.190
Welcome to Ubuntu 24.04.2 LTS (GNU/Linux 6.8.0-55-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu Oct 16 12:47:58 PM UTC 2025

System load:  0.0          Processes:      111
Usage of /:   40.8% of 14.66GB    Users logged in:  0
Memory usage: 14%          IPv4 address for ens18: 172.16.9.190
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

83 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

*** System restart required ***
root@eltex-practice2-pg1-v16:~#

```

```

eltex-pg1-v16@eltex-2025-autumn:~/.ssh$ cat id_rsa.pub
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQGBgQC8PGLHwB78/t3fz05XcJ0dLDCGZC00gy6DFV3poyHf
ceFPNCELTPzGU2DsFK3tznvESN3TqJA18sNHfnyoI9k+cxpAuD2oVF55j6kDYGSogJ7VPQX9SMPF0gB
OzraS1FyGdyaf9gJqeUD2F4k3rR0kRshBNCTQTVxwLbCTWx0B6QqBUP4LzZopTJNWHVutoqLDIFPLFHZ
0Rb4GeLuQbw0C91PASwHewEZr9VLX2Z79LVJ36GNEKB3oUNsq8ggtZ1Vn2ecPL1+XfmQ1CLDvFPHGJN
xZAzWbbJntszFEhJELsFhw6Zog6RRRTOMLJKfug06Q8U12zseEF4w4LwdqNhyNj6PnEZ/s/dfohvQq7at
9KcQuK9IXS91eh7mbvLN9YefBegEZAw4dcfncNXXU7V5JHN353b3n0APGJqf89UJjVkdSr3GzhtD2M
kpelHeHwewNzJN00CPHro+P/eUahX/9EFn0CyUJxcPnX6y+/74ZGqUY15ZMngKxzRAnSye0= eltex-p
g1-v16@eltex-2025-autumn
eltex-pg1-v16@eltex-2025-autumn:~/.ssh$

root@eltex-practice2-pg1-v16:~/.ssh# cat authorized_keys
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQGBgQC8PGLHwB78/t3fz05XcJ0dLDCGZC00gy6DFV3poyHf
ceFPNCELTPzGU2DsFK3tznvESN3TqJA18sNHfnyoI9k+cxpAuD2oVF55j6kDYGSogJ7VPQX9SMPF0gB
OzraS1FyGdyaf9gJqeUD2F4k3rR0kRshBNCTQTVxwLbCTWx0B6QqBUP4LzZopTJNWHVutoqLDIFPLFHZ
0Rb4GeLuQbw0C91PASwHewEZr9VLX2Z79LVJ36GNEKB3oUNsq8ggtZ1Vn2ecPL1+XfmQ1CLDvFPHGJN
xZAzWbbJntszFEhJELsFhw6Zog6RRRTOMLJKfug06Q8U12zseEF4w4LwdqNhyNj6PnEZ/s/dfohvQq7at
9KcQuK9IXS91eh7mbvLN9YefBegEZAw4dcfncNXXU7V5JHN353b3n0APGJqf89UJjVkdSr3GzhtD2M
kpelHeHwewNzJN00CPHro+P/eUahX/9EFn0CyUJxcPnX6y+/74ZGqUY15ZMngKxzRAnSye0= eltex-p
g1-v16@eltex-2025-autumn
root@eltex-practice2-pg1-v16:~/.ssh#

```

4. Создайте пользователя user1 при помощи команды useradd, укажите необходимость создания домашнего каталога и shell /bin/bash. Создайте пароль пользователю user1

```

root@eltex-practice2-pg1-v16:~# useradd -m -s /bin/bash user1
root@eltex-practice2-pg1-v16:~# passwd user1
New password:
Retype new password:
passwd: password updated successfully
root@eltex-practice2-pg1-v16:~#

```

5. Создайте пользователя user2 и user3 при помощи команды adduser

```

root@eltex-practice2-pg1-v16:~# adduser user2
info: Adding user 'user2' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group 'user2' (1002) ...
info: Adding new user 'user2' (1002) with group 'user2 (1002)' ...
info: Creating home directory '/home/user2' ...
info: Copying files from '/etc/skel' ...
New password:
Retype new password:
Sorry, passwords do not match.
passwd: Authentication token manipulation error
passwd: password unchanged
Try again? [y/N] y
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for user2
Enter the new value, or press ENTER for the default
  Full Name []: user2
    Room Number []: user2
      Work Phone []: 1
      Home Phone []: 1
        Other []: 1
Is the information correct? [Y/n] y
info: Adding new user 'user2' to supplemental / extra groups 'users' ...
info: Adding user 'user2' to group 'users' ...
root@eltex-practice2-pg1-v16:~#

```

```

root@eltex-practice2-pg1-v16:~# adduser user3
info: Adding user 'user3' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group 'user3' (1003) ...
info: Adding new user 'user3' (1003) with group 'user3 (1003)' ...
info: Creating home directory '/home/user3' ...
info: Copying files from '/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for user3
Enter the new value, or press ENTER for the default
  Full Name []: user3
    Room Number []: user3
    Work Phone []: 1
    Home Phone []: 1
    Other []: 1
Is the information correct? [Y/n] y
info: Adding new user 'user3' to supplemental / extra groups 'users' ...
info: Adding user 'user3' to group 'users' ...
root@eltex-practice2-pg1-v16:~#

```

6. Для пользователя user3 смените shell на /usr/sbin/nologin (man usermod), выполните вход под этим пользователем при помощи утилиты su, сначала без дополнительных параметров, затем с явным указанием shell /bin/bash в параметрах su. Выполните logout

```

root@eltex-practice2-pg1-v16:~# usermod --shell /usr/sbin/nologin user3
root@eltex-practice2-pg1-v16:~#

```

```

root@eltex-practice2-pg1-v16:~# su - user3
This account is currently not available.
root@eltex-practice2-pg1-v16:~# su -s /bin/bash user3
user3@eltex-practice2-pg1-v16:/root$ exit
exit
root@eltex-practice2-pg1-v16:~#

```

7. Создайте новую группу и добавьте её для всех пользователей user* как дополнительную, посмотрите список групп всех пользователей user*

```

root@eltex-practice2-pg1-v16:~# groupadd users0
root@eltex-practice2-pg1-v16:~# usermod -aG users0 user1
root@eltex-practice2-pg1-v16:~# usermod -aG users0 user2
root@eltex-practice2-pg1-v16:~# usermod -aG users0 user3

```

```

root@eltex-practice2-pg1-v16:~# id -Gn user1
user1 users0
root@eltex-practice2-pg1-v16:~# id -Gn user2
user2 users users0
root@eltex-practice2-pg1-v16:~# id -Gn user3
user3 users users0
root@eltex-practice2-pg1-v16:~#

```

8. Создайте каталог /opt/share и назначьте группу из предыдущего пункта его владельцем, установите на этот каталог бит SGID, права для группы rwx.

```

root@eltex-practice2-pg1-v16:~# mkdir -p /opt/share
root@eltex-practice2-pg1-v16:~# chgrp users0 /opt/share/
root@eltex-practice2-pg1-v16:~# chmod g+s /opt/share/
root@eltex-practice2-pg1-v16:~# chmod g+rwx /opt/
root@eltex-practice2-pg1-v16:~# chmod g+rwx /opt/share/
root@eltex-practice2-pg1-v16:~#

```

9. Для user1 задайте permanently umask, снимающий право чтения для «прочих»

```
root@eltex-practice2-pg1-v16:~# su - user1
user1@eltex-practice2-pg1-v16:~$ ls -la
.  ..  .bash_logout  .bashrc  .profile
user1@eltex-practice2-pg1-v16:~$ nano .bashrc
user1@eltex-practice2-pg1-v16:~$ source .bashrc
user1@eltex-practice2-pg1-v16:~$
```

Задание umask для user1 в файле .bashrc

10. Создайте каждым из пользователей новые файлы в каталоге /opt/share, удалите файлы созданные другими пользователями

```
root@eltex-practice2-pg1-v16:~# su -s /bin/bash user3
user3@eltex-practice2-pg1-v16:/root$ cd ..
user3@eltex-practice2-pg1-v16:/opt$ cd share/
user3@eltex-practice2-pg1-v16:/opt/share$ touch 3.txt
user3@eltex-practice2-pg1-v16:/opt/share$
```

```
root@eltex-practice2-pg1-v16:~# su -s /bin/bash user2
user2@eltex-practice2-pg1-v16:/root$ cd ..
user2@eltex-practice2-pg1-v16:/opt$ cd share/
user2@eltex-practice2-pg1-v16:/opt/share$ touch 2.txt
user2@eltex-practice2-pg1-v16:/opt/share$ rm 3.txt
user2@eltex-practice2-pg1-v16:/opt/share$ ls
2.txt
user2@eltex-practice2-pg1-v16:/opt/share$
```

```
root@eltex-practice2-pg1-v16:~# su user1
user1@eltex-practice2-pg1-v16:/root$ cd ..
user1@eltex-practice2-pg1-v16:/opt$ cd share/
user1@eltex-practice2-pg1-v16:/opt/share$ touch 1.txt
user1@eltex-practice2-pg1-v16:/opt/share$ rm 2.txt
user1@eltex-practice2-pg1-v16:/opt/share$ ls
1.txt
user1@eltex-practice2-pg1-v16:/opt/share$
```

11. Повторите предыдущий пункт, предварительно установив sticky bit на каталоге /opt/share

```
root@eltex-practice2-pg1-v16:~# chmod +t opt/share/
root@eltex-practice2-pg1-v16:~# su user1
user1@eltex-practice2-pg1-v16:/opt$ cd share/
user1@eltex-practice2-pg1-v16:/opt/share$ rm 1.txt
user1@eltex-practice2-pg1-v16:/opt/share$ touch 1.txt
Command 'touch' not found, did you mean:
  command 'touch' from deb coreutils (9.4-3ubuntu6.1)
Try: apt install <deb name>
user1@eltex-practice2-pg1-v16:/opt/share$ touch 1.txt
user1@eltex-practice2-pg1-v16:/opt/share$ ls
1.txt
user1@eltex-practice2-pg1-v16:/opt/share$
```



```

root@eltex-practice2-pg1-v16:/# su user2
user2@eltex-practice2-pg1-v16:/$ cd opt/
user2@eltex-practice2-pg1-v16:/opt$ cd share/
user2@eltex-practice2-pg1-v16:/opt/share$ rm 1.txt
rm: remove write-protected regular empty file '1.txt'?
user2@eltex-practice2-pg1-v16:/opt/share$ touch 2.txt
Command 'touch' not found, did you mean:
  command 'touch' from deb coreutils (9.4-3ubuntu6.1)
Try: apt install <deb name>
user2@eltex-practice2-pg1-v16:/opt/share$ touch 2.txt
user2@eltex-practice2-pg1-v16:/opt/share$ ls
1.txt 2.txt
user2@eltex-practice2-pg1-v16:/opt/share$

```

```

root@eltex-practice2-pg1-v16:/# su -s /bin/bash user3
user3@eltex-practice2-pg1-v16:/$ cd opt/
user3@eltex-practice2-pg1-v16:/opt$ cd share/
user3@eltex-practice2-pg1-v16:/opt/share$ rm 1.txt
rm: remove write-protected regular empty file '1.txt'?
user3@eltex-practice2-pg1-v16:/opt/share$ rm 2.txt
rm: cannot remove '2.txt': Operation not permitted
user3@eltex-practice2-pg1-v16:/opt/share$ touch 3.txt
user3@eltex-practice2-pg1-v16:/opt/share$ ls
1.txt 2.txt 3.txt
user3@eltex-practice2-pg1-v16:/opt/share$

```

12. Разрешите user1 выполнять привилегированную команду dmesg при помощи команды sudo, а user2 – при помощи скрипта на языке bash с установленным флагом SUID

```

GNU nano 7.2 /etc/sudoers.tmp

# User alias specification

# Cmnd alias specification

# User privilege specification
root    ALL=(ALL:ALL) ALL

# Members of the admin group may gain root privileges
%admin   ALL=(ALL) ALL

# Allow members of group sudo to execute any command
%sudo   ALL=(ALL:ALL) ALL

%user1  ALL=(ALL) NOPASSWD: /bin/dmesg

```

```

GNU nano 7.2 dng.sh *
#!/bin/bash
/usr/bin/dmesg "$@"

```

```

[88542.153218] audit: type=1400 audit(1760603259.603:282): apparmor="DENIED" operation="mknod" class="file" profile="ubuntu_pro_apt_news" name="/usr/lib/python3/dist-packages/uaclient/api/u/pro/status/__pycache__/__init__.cpython-312.pyc.140688616457248" pid=35854 comm="python3" requested_mask="c" denied_mask="c" fsuid=0 ouid=0
[88542.300964] audit: type=1400 audit(1760603259.751:283): apparmor="DENIED" operation="mknod" class="file" profile="ubuntu_pro_apt_news" name="/usr/lib/python3/dist-packages/uaclient/api/u/pro/status/is_attached/__pycache__/__init__.cpython-312.pyc.140688616480624" pid=35854 comm="python3" requested_mask="c" denied_mask="c" fsuid=0 ouid=0
[88542.339233] audit: type=1400 audit(1760603259.789:284): apparmor="DENIED" operation="mknod" class="file" profile="ubuntu_pro_esm_cache" name="/usr/lib/python3/dist-packages/uaclient/__pycache__/actions.cpython-312.pyc.136541546612400" pid=35855 comm="python3" requested_mask="c" denied_mask="c" fsuid=0 ouid=0
root@eltex-practice2-pg1-v16:~# chmod u+s dng.sh
root@eltex-practice2-pg1-v16:~# chmod +x dng.sh
root@eltex-practice2-pg1-v16:~#

```

13. Для всех пользователей user* задайте время действия пароля – 10 дней.

```
root@eltex-practice2-pg1-v16:~# chage -M 10 user1
root@eltex-practice2-pg1-v16:~# chage -M 10 user2
root@eltex-practice2-pg1-v16:~# chage -M 10 user3
root@eltex-practice2-pg1-v16:~#
```

14. Отредактируйте файл /etc/motd, вписав туда свое имя и фамилию

```
root@eltex-practice2-pg1-v16:~# nano /etc/motd
root@eltex-practice2-pg1-v16:~# cat /etc/motd
Коротков Александр
root@eltex-practice2-pg1-v16:~#
```

15. Создайте копию содержимого каталога /etc в каталог /root/etc_backup при помощи программы rsync

```
root@eltex-practice2-pg1-v16:~# rsync /etc/* /root/etc_backup
skipping directory ModemManager
skipping directory PackageKit
skipping directory UPower
skipping directory X11
skipping directory alternatives
```

16. Заархивируйте содержимое каталога /root/etc_backup архиватором tar, используйте алгоритмы сжатия gzip, bzip2, 7zip, сравните размеры полученных файлов

```
root@eltex-practice2-pg1-v16:~# tar -czf gzip_back.tar.gz root/etc_backup
root@eltex-practice2-pg1-v16:~# tar -cjf bzip_back.tar.gz root/etc_backup
root@eltex-practice2-pg1-v16:~# tar -cvf szip_back.tar root/etc_backup
```

```
root@eltex-practice2-pg1-v16:~# 7z a szip_back.tar.7z szip_back.tar
7-Zip 23.01 (x64) : Copyright (c) 1999-2023 Igor Pavlov : 2023-06-20
64-bit locale=C.UTF-8 Threads:2 OPEN_MAX:1024

Scanning the drive:
1 file, 327680 bytes (320 KiB)

Creating archive: szip_back.tar.7z

Add new data to archive: 1 file, 327680 bytes (320 KiB)

Files read from disk: 1
Archive size: 71087 bytes (70 KiB)
Everything is Ok
root@eltex-practice2-pg1-v16:~#
```

```
root@eltex-practice2-pg1-v16:~# du -sh gzip_back.tar.gz
84K    gzip_back.tar.gz
root@eltex-practice2-pg1-v16:~# du -sh bzip_back.tar.gz
76K    bzip_back.tar.gz
root@eltex-practice2-pg1-v16:~# du -sh szip_back.tar.7z
72K    szip_back.tar.7z
root@eltex-practice2-pg1-v16:~#
```

17. Отредактируйте файл /etc/motd, вписав туда текущую дату и время, синхронизируйте каталог /root/etc_backup с каталогом /etc при помощи rsync, добавьте файл motd в архив, сжатый gzip

```

root@eltex-practice2-pg1-v16:/# nano /etc/motd
root@eltex-practice2-pg1-v16:/# cat /etc/motd
Коротков Александр
16.10.2025 23:10
root@eltex-practice2-pg1-v16:/# rsync -a /root/etc_backup /etc
root@eltex-practice2-pg1-v16:/# █

```

Невозможно добавить файл в архив, не пересоздавая его.

18. Сравните содержимое архива, упакованного bzip2 с содержимым каталога /root/etc_backup

```

root@eltex-practice2-pg1-v16:/# tar -xjf bz1p_back.tar.gz -C bz
root@eltex-practice2-pg1-v16:/# diff /bz/root/etc_backup/ /root/etc_backup/
root@eltex-practice2-pg1-v16:/# █

```

19. Распакуйте архивы etc_backup, упакованные gzip и 7zip в каталоги /root/etc_backup_gzip и /root/etc_backup_7zip, сравните программой diff файлы motd в этих каталогах.

```

root@eltex-practice2-pg1-v16:/# 7z x sz1p_back.tar.7z -o/root/etc_backup_7zip/
7-Zip 23.01 (x64) : Copyright (c) 1999-2023 Igor Pavlov : 2023-06-20
64-bit locale=C.UTF-8 Threads:2 OPEN_MAX:1024

Scanning the drive for archives:
1 file, 71087 bytes (70 KiB)

Extracting archive: sz1p_back.tar.7z
--
Path = sz1p_back.tar.7z
Type = 7z
Physical Size = 71087
Headers Size = 130
Method = LZMA2:384k
Solid = -
Blocks = 1

Everything is Ok

Size:          327680
Compressed: 71087
root@eltex-practice2-pg1-v16:/# █

```

```

root@eltex-practice2-pg1-v16:/# tar -xf /root/etc_backup_7zip/sz1p_back.tar -C /
root@eltex-practice2-pg1-v16:/# █

```

```

root@eltex-practice2-pg1-v16:/# diff /root/etc_backup_7zip/res/root/etc_backup/m
otd /root/etc_backup_gzip/root/etc_backup/motd
root@eltex-practice2-pg1-v16:/# █

```